Midterm I Retake Solution.

Q1 Answer:

$$\bar{F}_1 = 5.3922 \times 10^{14}\, a_x + 1.07844 \times 10^{15}\, a_y \quad N$$

$$|\bar{F}_1| = F_1 = 1.2043 \times 10^{15}\, N$$

$$\bar{F}_2 = -7.9342 \times 10^{14}\, a_x + 2.542 \times 10^{14}\, a_y$$

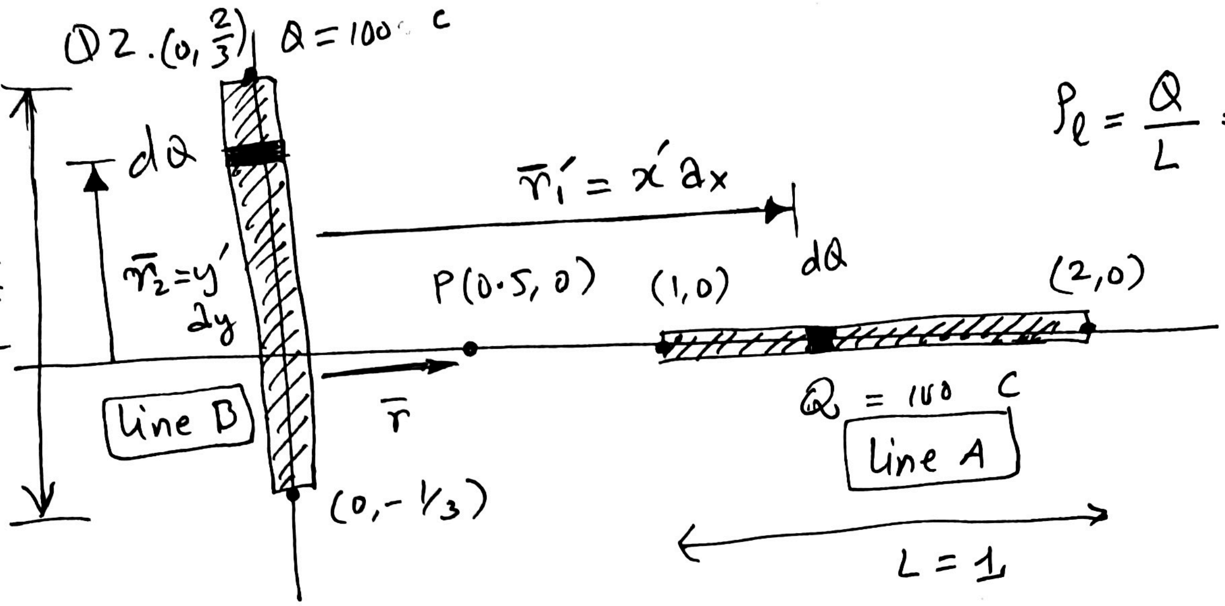
$$|\bar{F}_2| = F_2 = 8.318 \times 10^{14}\, N$$

$$\bar{F}_3 = 2.342 \times 10^{14}\, a_x - 1.33244 \times 10^{15}\, a_y$$

$$|\bar{F}_3| = F_3 = 1.355 \times 10^{15}\, N$$

F_3 is the largest.

Q2.



$$\rho_l = \frac{Q}{L} = \frac{Q}{1} = Q$$

Consider line A

$$\bar{E}_A = \frac{1}{4\pi\epsilon_0}\int \frac{\rho_l\, dl\, (\bar{r} - \bar{r}_1')}{|\bar{r} - \bar{r}_1|^3}$$

$dl = dx'$, $\rho_l = Q$,

$\bar{r} = 0.5a_x$,

$\bar{r}_1' = x'a_x$, $(1 \leq x' \leq 2)$

$$\bar{E}_A = \frac{Q}{4\pi\epsilon_0}\int_{x'=1}^{2} \frac{(0.5 - x')\,dx'\, a_x}{|(0.5 - x')|^3}$$

for $Q = 100$ C (nano is dropped for convenience), we get $\bar{E}_A = 1.2 \times 10^{12}$ N/C.

$$= \frac{Q}{4\pi\epsilon_0} a_x \int \frac{(0.5 - x')\,dx'}{(x' - 0.5)^3}$$

$\boxed{x' > 0.5}$ let $u = x' - 0.5$, $du = dx'$

$$= -\frac{Q\, a_x}{4\pi\epsilon_0}\int_{x'=1}^{2} \frac{dx'}{(x' - 0.5)^2}$$

$$= -\frac{Q\, a_x}{4\pi\epsilon_0}\int_{u=0.5}^{1.5} \frac{du}{u^2} = \frac{-Q\, a_x}{4\pi\epsilon_0} \left. \frac{u^{-2+1}}{-2+1} \right|_{0.5}^{1.5}$$

$$= \frac{Q\, a_x}{4\pi\epsilon_0}\left(\frac{1}{1.5} - \frac{1}{0.5}\right)$$

$$= \frac{Q\, a_x}{4\pi\epsilon_0}\left(\frac{2}{3} - 2\right)$$

$$\bar{E}_A = \frac{Q\, a_x}{2\pi\epsilon_0}\left(-\frac{2}{3}\right) = \frac{-Q\, a_x}{3\pi\epsilon_0}$$

Consider line B

$$\bar{E}_B = \frac{1}{4\pi\epsilon_0}\int \frac{Q\, dy' (\bar{r} - \bar{r}_2')}{|\bar{r} - \bar{r}_2'|^3}$$

$$\bar{r}_2' = y' a_y \qquad \left(-\frac{1}{3} \le y' \le \frac{2}{3}\right)$$

$$\bar{E}_B = \frac{Q}{4\pi\epsilon_0}\int_{y'=-\frac{1}{3}}^{y'=2/3} \frac{dy' (0.5 a_x - y' a_y)}{(0.5^2 + y'^2)^{3/2}}$$

$$\boxed{\begin{aligned} \int \frac{x\,dx}{(a^2+x^2)^{3/2}} &= -\frac{1}{\sqrt{a^2+x^2}} + C \\ \int \frac{dx}{(a^2+x^2)^{3/2}} &= \frac{x}{a^2\sqrt{a^2+x^2}} + C \end{aligned}}$$

$$\bar{E}_B = \frac{Q}{4\pi\epsilon_0}\left[\frac{0.5 a_x y'}{(0.5)^2\sqrt{0.5^2 + y'^2}}\Bigg|_{y'=-\frac{1}{3}}^{y'=2/3} + \frac{a_y}{\sqrt{0.5^2 + y'^2}}\Bigg|_{y'=-\frac{1}{3}}^{y'=\frac{2}{3}} \right]$$

$$= \frac{Q}{4\pi\epsilon_0}\left[\frac{2a_x\left(\frac{2}{3}\right)}{\sqrt{0.5^2 + (2/3)^2}} - \frac{2a_x\left(-\frac{1}{3}\right)}{\sqrt{0.5^2 + (1/3)^2}} + \frac{a_y\left(\frac{2}{3}\right)}{\sqrt{0.5^2 + (2/3)^2}} - \frac{a_y\left(-\frac{1}{3}\right)}{\sqrt{0.5^2 + (1/3)^2}} \right]$$

Net Electric field is $\bar{E} = \bar{E}_A + \bar{E}_B$.

Consider line A

$$V_A = \frac{1}{4\pi\epsilon_0}\int \frac{\rho_\ell \, dl'}{|\bar{r}-\bar{r}'|} = \frac{1}{4\pi\epsilon_0}\int \frac{Q\,dx'}{|0.5 - x'|}$$

$$= \frac{1}{4\pi\epsilon_0}\int_{x'=1}^{2} \frac{Q\,dx'}{(x'-0.5)}$$

Compare the two, absolute operator is removed and x' appears first because $x' > 0.5$.

$$= \frac{Q}{4\pi\epsilon_0} \ln(x'-0.5)\Big|_1^2$$

$$= \frac{Q}{4\pi\epsilon_0}\Big(\ln(0.5) - \ln(1.5)\Big)$$

The total potential is the sum of V_A & V_B.

Consider line B

$$V_B = \frac{1}{4\pi\epsilon_0}\int_{y'=-\frac{1}{3}}^{2/3} \frac{Q\,dy'}{(0.5^2+y'^2)^{\frac{1}{2}}}$$

$$\boxed{\int \frac{dx}{\sqrt{a^2+x^2}} = \tanh^{-1}\left(\frac{x}{\sqrt{a^2+x^2}}\right) + c}$$

$$V_B = \frac{Q}{4\pi\epsilon_0} \tanh^{-1}\left(\frac{y'}{\sqrt{0.5^2+y'^2}}\right)\Big|_{-1/3}^{2/3}$$

$$= \frac{Q}{4\pi\epsilon_0}\left(\tanh^{-1}\left(\frac{2/3}{\sqrt{0.25+(2/3)^2}}\right) - \tanh^{-1}\left(\frac{-1/3}{\sqrt{0.25+(1/3)^2}}\right)\right)$$

Q3. Answer/Solution.

(a) $\rho_v(r') = \rho_0\left(1 - \frac{r'}{a}\right) \qquad 0 < r' < a$

$$Q = \int \rho_v \, dv = \int_{r=0}^{a}\int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi} \rho_0\left(1-\frac{r}{a}\right) r^2 \sin\theta \, dr\, d\theta\, d\phi$$

$$Q = \frac{\pi \rho_0 a^3}{3}.$$

(b) $\oint \bar{D}\cdot d\bar{S} = Q_{enc}.$

or $\oint \bar{E}\cdot d\bar{S} = \frac{Q_{enc}}{\epsilon_0}$

Solving this, we obtain. $\bar{E} = \frac{\rho_0 r}{\epsilon_0}\left(\frac{1}{3} - \frac{r}{4a}\right)\bar{a}_r$

$0 < r < a$

For outside; $\bar{E} = \frac{\rho_0 a^3}{12\,\epsilon_0 r^2}\bar{a}_r \; (r > a).$

(c) $V = -\int \bar{E}\cdot d\bar{l}$

Outside: $r > a$.

$$V = -\int_{\infty}^{r} \frac{\rho_0 a^3}{12\,\epsilon_0 r^2}\, dr = -\frac{\rho_0 a^3}{12\,\epsilon_0}\left.\frac{r^{-2+1}}{-2+1}\right|_{\infty}^{r}$$

$$= \left.\frac{\rho_0 a^3}{12\,\epsilon_0 r}\right|_{\infty}^{r}$$

$$= \frac{\rho_0 a^3}{12\,\epsilon_0}\left(\frac{1}{r} - \frac{1}{\infty}\right)$$

$$\boxed{V(r) = \frac{\rho_0 a^3}{12\epsilon_0 r}} \qquad (r > a)$$

(b) inside $\rho < a$

$$V = -\int_{\infty}^{a} \overline{E}_{outside} \cdot \overline{dl} - \int_{a}^{r} \overline{E}_{inside} \cdot \overline{dl}$$

$$= \frac{\rho_0 a^3}{12\, \epsilon_0 r}\Big|_{r=a} - \int_a^r \left(\frac{\rho_0 r'}{3\, \epsilon_0} - \frac{\rho_0 r'^2}{4 a \epsilon_0} \right) dr'$$

$$= \frac{\rho_0 a^2}{12\, \epsilon_0} - \frac{\rho_0}{3\epsilon_0}\frac{r'^2}{2} + \frac{\rho_0 r'^3}{12 a \epsilon_0}\Big|_a^r$$

$$= \frac{\rho_0 a^2}{12\, \epsilon_0} - \frac{\rho_0 (r^2 - a^2)}{6\epsilon_0} + \frac{\rho_0 (r^3 - a^3)}{12 a \epsilon_0}.$$

$$= \frac{\rho_0 r^3}{12 a \epsilon_0} - \frac{\rho_0 (r^2 - a^2)}{6\epsilon_0}.$$